

DETAILED PROJECT REPORT (DPR) - V3

MY JEEVANDHARA ECO-DATA CENTER COMPLEX

*Bulletproof EIA Environmental Clearance Architecture & Operational Blueprint | Location:
Kapuluppada, Visakhapatnam*

1. EXECUTIVE SUMMARY & STRATEGIC MANDATE

Traditional data centers present an critical threat to regional water security. High-performance computational infrastructure demands massive, continuous thermal cooling loops. In conventional layouts, this requires millions of liters of pristine groundwater extracted daily via evaporative cooling networks, triggering localized water table collapse and environmental conflict with adjacent agricultural or residential zones.

The My Jeevandhara Sovereign Resolution: This Detailed Project Report establishes a standalone 100-Acre Hyper-Scale Eco-Data Center Facility in Visakhapatnam operating at **100% Net Water Neutrality** and **Zero Groundwater Dependency**. By deploying our infrastructure matrix, the complex eliminates external water table pressure, actively harvests marine moisture, and recycles computer-generated thermal exhaust to sustain its cooling lines.

2. CORE SITE ARCHITECTURE & LAND SPECIFICATIONS

The project occupies a standalone **100-Acre site in the Kapuluppada IT SEZ hill buffer zone, Visakhapatnam**. This plot is decoupled from the primary 1000-acre Sovereign Akanda Bharat Township, selected specifically for its strategic elevation (preventing flash flooding risks), quick proximity to subsea internet cable landing corridors, and dense marine atmospheric humidity streams.

ZONE ALLOCATION	ACREAGE	PRIMARY INFRASTRUCTURE ELEMENTS
Zone A: Computational Core	30 Acres	4x Hyper-scale Data Center Server Blocks, Jarvis AI Command Hub, Power Substations.
Zone B: Industrial Water Engine	35 Acres	Industrial Air-to-Water (AWG) Manufacturing Farms, Dual-Ion Shield Filtration Building.
Zone C: Thermal Recycling & Desalination	20 Acres	Low-Pressure Vacuum Distillation Loops, Heat Exchanger Terminals, Solar Stills Co-Gen Plant.
Zone D: Sovereign Green Energy Grid	15 Acres	Bifacial Rooftop Solar Arrays, Energy Storage Systems (ESS), Peripheral Aeroleaf Wind Corridors.

3. TECHNICAL WATER RESOLUTION & THE 4-PILLAR RECIPIENT MATRICES

By integrating advanced winter cooling protocols, the facility systematically resolves the three major municipal objections (Water, Power, Public) and satisfies the most stringent regulatory criteria:

- **1. WATER (Zero-Groundwater Infrastructure):** Absolute zero groundwater extraction policy. By utilizing advanced automated atmospheric condensation combined with ambient free cooling mechanics, the facility achieves up to **4 full months (120 days of winter) of zero active water consumption** for thermal management.
- **2. POWER (Efficiency Multiplier):** The system extracts massive seasonal cold atmospheric gradients to achieve a **40% structural energy saving** during winter operation. This successfully lowers the peak regional grid load by **2 MW** continuous.
- **3. PUBLIC (Community Resource Security):** No industrial borewells are permitted on site. The 2 MW electrical load reduction directly frees up grid power capacity equivalent to the power demands of ****500 local rural households****, turning the data center into a resource-positive community asset.
- **4. COMPLIANCE (Sovereign Engineering Standards):** Complete validation across global and national benchmarks. The facility complies fully with **ASHRAE thermal envelopes (18°C – 27°C)**, MoEFCC guidelines, and APPCB industrial criteria.

4. STANDALONE BULLETPROOF REGULATORY CLEARANCE ADDITIONS

To guarantee immediate, uncompromised approvals during environmental impact assessment (EIA) hearings and to address technical queries from APPCB/MoEFCC officers, the following engineered protocols are integrated into the facility's water infrastructure chapter:

A. Monsoon Backup Plan (High Humidity Operational Adaptation)

"During periods of heavy monsoon downpours or sustained high-humidity windows where open-air thermal condensation limits stabilize, the computational cooling system automatically shifts into an advanced closed-loop chiller configuration utilizing treated Sewage Treatment Plant (STP) greywater loops. This configuration completely isolates the infrastructure from local municipal water bodies, maintaining absolute zero groundwater dependency across all seasons, weather conditions, or environmental extremes."

B. Measurement, Verification & SCADA Enforcement Plan

"To provide auditable, real-time proof of zero-extraction status to regulatory authorities, an integrated array of industrial smart water inline meters coupled with a centralized SCADA system will record every cubic meter of water sourced and processed daily. An automated, tamper-proof monthly monitoring report will be compiled and submitted directly to the APPCB, confirming complete compliance with the zero groundwater extraction mandate."

**5. RISK MITIGATION & ENVIRONMENTAL IMPACT ASSESSMENT (EIA)
STATUS MATRIX**

EIA RISK PARAMETER EVALUATED	MY JEEVANDHARA ENGINEERED COUNTERMEASURE	REGULATORY CLEARANCE STATUS
Groundwater Depletion	100% Zero extraction mandate + 120 days seasonal free ambient cooling cycles.	✓ COVERED / AUDIT READY
Local Power Grid Overload	2 MW structural winter energy savings via low-temperature thermal deltas.	✓ COVERED / AUDIT READY
Environmental & Thermal Discharges	100% Closed-Loop Zero Liquid Discharge (ZLD) complying with ASHRAE + 120-day free cycles.	✓ COVERED / AUDIT READY
Monsoon Climatic Deviations	Closed-loop auxiliary chiller shifting using secondary treated STP greywater streams.	✓ COVERED / AUDIT READY
Data Manipulation / Compliance Faking	Real-time industrial SCADA measurement loops paired with automated monthly APPCB direct filing.	✓ COVERED / AUDIT READY

6. SEASONAL THERMODYNAMICS: WINTER OPTIMIZATION MATRIX

The complex converts winter climatic conditions (19°C - 28°C with 72% - 76% relative humidity) into an active operational efficiency asset:

- **Dew-Point Phase Optimization:** As ambient coastal air cools, it reaches its liquid dew point with minimal mechanical compression, triggering a **30% reduction in electrical condensation loading**.
- **Maximization of Thermal Delta (ΔT):** While external air cools, computing core nodes maintain stable heat profiles (35°C - 55°C). This maximizes the thermal gradient gap between cold inlet fluids and heated outlets, accelerating the low-pressure vacuum distillation cycles.
- **Surroundings Community Protection Line:** Despite cold seasonal dry spells, the facility maintains its commitment to local populations, routing **150,000 Liters of Fresh Drinking Water Daily** free of cost to adjacent farming pockets via a dedicated exterior pipeline distribution network.

7. TOTAL INITIAL CAPEX BUDGET SUMMARY

INFRASTRUCTURE BLOCK COMPONENT	ENGINEERING SPECIFICATIONS & CORE ASSETS	ESTIMATED COST (INR CRORES)
Land & Core Substructure	100 Acres grading, heavy leveling, civil foundation, security parameter mesh.	₹120.00 Cr
Computational Infrastructure	Server racks, network switching, fiber integration, Jarvis AI core nodes.	₹650.00 Cr
Industrial AWG Array Blocks	High-capacity coastal moisture condensation intake terminals.	₹180.00 Cr
Dual-Ion Shield Building	Automated Copper/Silver inline ionization channels and filtration banks.	₹35.00 Cr
Vacuum Co-Gen & SCADA Plant	Low-pressure thermal distillers, SCADA tracking sensors, greywater recovery loops.	₹95.00 Cr
Sovereign Solar Grid (35 MW)	High-efficiency bifacial monocrystalline arrays + local ESS battery banks.	₹140.00 Cr
Community Outflow Network	15km peripheral distribution pipe network and smart free-dispensing kiosks.	₹15.00 Cr
TOTAL INITIAL CAPITAL EXPEDXITURE	Turnkey Operational Construction (0 - 24 Months Timeline)	₹1,235.00 Crores

8. HUMAN CAPITAL RESOURCE EQUALITY (50% MEN / 50% WOMEN MANDATE)

In compliance with the foundational human resource protocols of the My Jeevandhara infrastructure, the complex enforces a non-negotiable ****50% Male / 50% Female gender parity framework**** across all operational tiers:

EMPLOYMENT TIER	MALE STAFF	FEMALE STAFF	PRIMARY FUNCTIONAL MANDATE
Tier 1: AI & Data Scientists	20 Personnel	20 Personnel	Jarvis AI operations, algorithm logic, computational workload routing.
Tier 2: Thermodynamic Engineers	35 Personnel	35 Personnel	Vacuum loops management, AWG arrays tracking, SCADA audit logs.
Tier 3: Grid & Solar Operators	25 Personnel	25 Personnel	35 MW Solar generation monitoring, ESS charge loops synchronization.
Tier 4: Security & Support Crews	40 Personnel	40 Personnel	Physical perimeter protection, hardware line routing, facility upkeep.
TOTAL INFRASTRUCTURE STAFF	120 Men	120 Women	240 Full-Time Ground-Zero to High-Level Specialists

9. OPEX & REVENUE CONCLUSION

The standalone facility incurs an ****Annual Operating Expense (Opex) of ₹42.00 Crores / Year**** and generates an ****Annual Revenue of ₹185.00 Crores / Year**** through sovereign cloud leasing and high-performance data traffic routing. This delivers a ****Net Annual Profit Surplus of ₹143.00 Crores / Year****, yielding an annual ROI of ****11.57%**** and securing a complete capital payback breakeven point within ****8.5 Years****.